

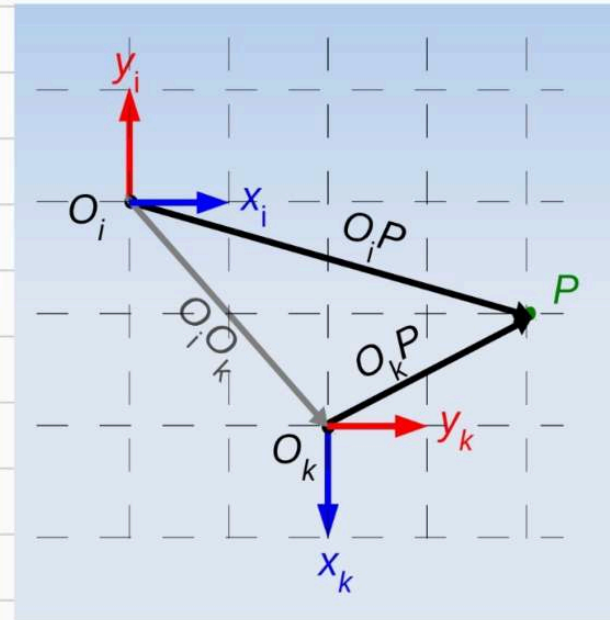
Mouvements, changements de coordonnées

Exercice, calculer ${}^i P = {}^i R_R \cdot {}^R P + {}^i O_R$

$${}^i R_R = \begin{bmatrix} \overset{0}{\cos \theta} & \overset{-1}{-\sin \theta} & 0 \\ \overset{1}{\sin \theta} & \overset{0}{\cos \theta} & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \theta = \pi/2$$

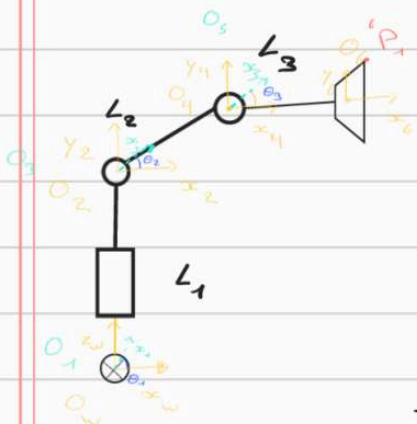
$${}^R P = \begin{bmatrix} -1 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

$${}^i O_R = \begin{bmatrix} 2 \\ -2 \\ 0 \\ 1 \end{bmatrix}$$



$$\left. \begin{aligned} x_R &= -y_i \\ y_R &= x_i \\ z_R &= z_i \end{aligned} \right\} \begin{aligned} \vec{x}_R &= \vec{x}_i \cdot 0 - \vec{y}_i + \vec{z}_i \cdot 0 \\ \vec{y}_R &= \vec{x}_i + \vec{y}_i \cdot 0 + \vec{z}_i \cdot 0 \\ \vec{z}_R &= \vec{x}_i \cdot 0 + \vec{y}_i \cdot 0 + \vec{z}_i \end{aligned} \quad \left. \begin{aligned} & \text{Permet de trouver} \\ & {}^i R_R \end{aligned} \right\}$$

On a donc ${}^i T_R = \left(\begin{array}{ccc|c} 0 & -1 & 0 & 2 \\ 1 & 0 & 0 & -2 \\ 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 1 \end{array} \right)$



On cherche à exprimer les coordonnées de l'effecteur dans le repère monde

$${}^w T_1 = \text{ROT}/z_w (\theta_1)$$

$${}^1 T_2 = \text{TRANS} (0, 0, L_1)$$

$${}^2 T_3 = \text{ROT}/y_2 (\theta_2)$$

$${}^3 T_4 = \text{TRANS} (0, 0, L_2)$$

$${}^4 T_5 = \text{ROT}/y_4 (\theta_3)$$

$${}^5 T_6 = \text{TRANS} (0, 0, L_3)$$

Par trouver les coordonnées de ${}^w P_1$ on a

$$\begin{bmatrix} {}^w P_1 \\ 1 \end{bmatrix} = {}^w T_6 \begin{bmatrix} {}^6 P_1 \\ 1 \end{bmatrix}$$

Calcul de ${}^wT_1 = \text{ROT}/z_w(\theta_1)$

$$\begin{pmatrix} \cos \theta_1 & -\sin \theta_1 & 0 \\ \sin \theta_1 & \cos \theta_1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Calcul de ${}^2T_3 = \text{ROT}/y_2(\theta_2)$

$$\begin{pmatrix} \cos \theta_2 & 0 & \sin \theta_2 \\ 0 & 1 & 0 \\ -\sin \theta_2 & 0 & \cos \theta_2 \end{pmatrix}$$

Calcul de ${}^4T_5 = \text{ROT}/y_4(\theta_3)$

$$\begin{pmatrix} \cos \theta_3 & 0 & \sin \theta_3 \\ 0 & 1 & 0 \\ -\sin \theta_3 & 0 & \cos \theta_3 \end{pmatrix}$$

Calcul de ${}^1T_2 = \text{TRANS}(0,0,L)$

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